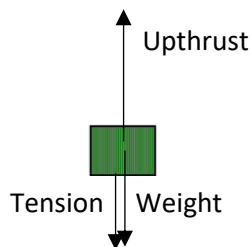


moving in the negative direction, which is downwards.

2. Speed of the balloon is zero [because initial total momentum = 0].

2(a) (i)



(a)(ii) Total upward forces = total downward forces [CONCEPT]

$$T + mg = U$$

$$kx + mg = U$$

$$L = \frac{V\rho g - mg}{k} = \frac{\frac{2}{3}\left(\frac{5.0}{650}\right)(1000)(9.81) - (5.0)(9.81)}{160} \quad \text{where } x = L$$

$$L = 7.86 \times 10^{-3} \text{ m}$$

(b)(i) Taking pivot at A, total clockwise moment = total anti-clockwise moment, [CONCEPT]

$$\left(\frac{AC}{2} \cos 65^\circ\right)(1200) + (AC \cos 65^\circ)(2000) = T(AB)$$

$$\left(\frac{AC}{2} \cos 65^\circ\right)(1200) + (AC \cos 65^\circ)(2000) = T\left(\frac{3}{4}AC\right)$$

$$T = 1.47 \times 10^3 \text{ N}$$

(b)(ii) Since tension exerted on the boom is at an angle, there will be a horizontal component. [APPLICATION]

For the boom to be in equilibrium, there must be a horizontal force to counter this force. [CONCEPT: FORCES IN EQUILIBRIUM] Therefore, the force at the hinge is not vertical but at an angle to provide a horizontal component to counter the horizontal component of the tension. [CONCLUSION]

2(c) Using $n\lambda = nRT$ [CONCEPT]